Interface summary

31 interfaces and 10 type aliases from 10 files, in 7 groups of related interfaces. Inherited members are listed under the interface that declares them. Generated 2026-11-11 by npm run summary.

RectangleOps group

Point geometry.ts

A location, in the coordinate space its use states.

Used by RectangleOps; Shape, InkStroke, Circle, ShapeLocatorQueries (page 2); EventDistributor (page 3).

```
readonly x: number
readonly y: number
```

Rectangle geometry.ts

An axis-aligned rectangle covering the half-open region [x, x + width) × [y, y + height).

Used by RectangleOps; Shape, ShapeLocatorQueries (page 2); LayoutObject (page 3).

```
readonly x: number
readonly y: number
readonly width: number
readonly height: number
```

RectangleOps geometry.ts

Operations on rectangles, following Rectangle's half-open semantics. cut, splitHoriz, splitVert, inset and place are each defined exactly by their reference functions in geometry.vocab.ts.

```
contains(r: Rectangle, p: Point): boolean
```
Whether p lies in r.

```
intersects(a: Rectangle, b: Rectangle): boolean
```
Whether a and b overlap.

```
cut(r: Rectangle, edge: Edge, size: number): [Rectangle,
  Rectangle]
```
Cuts a strip of the given size off one edge of r.

```
splitHoriz(r: Rectangle, ...weights: number[]):
  Rectangle[]
```
Splits r left to right into one rectangle per weight, each as wide as its share of r's width, and as tall as r. splitHoriz(r, 5, 10, 5) gives a quarter, a half and a quarter.

```
splitVert(r: Rectangle, ...weights: number[]):
  Rectangle[]
```
Splits r top to bottom into one rectangle per weight, each as tall as its share of r's height, and as wide as r.

RectangleOps, continued

```
inset(r: Rectangle, d: number): Rectangle
```
r shrunk by d on every side.

```
place(r: Rectangle, size: { readonly width: number;
  readonly height: number }, alignX: Align, alignY:
  Align): Rectangle
```
A rectangle of the given size placed within r, aligned along each axis at the start, the center (rounded down to a whole unit) or the end.

Viewport geometry.ts

A 2D similarity transform (uniform scale, rotation, translation) from canonical coordinates to view coordinates (see the top of this file).

Used by View (page 4).

```
readonly scale: number
```
View units per canonical unit.

```
readonly rotation: number
```
Radians.

```
readonly tx: number
readonly ty: number
```

Types

```
type Edge = "top" | "bottom" | "left" | "right"
```
A side of a rectangle.

Used by RectangleOps.

```
type Align = "start" | "center" | "end"
```
Where a smaller rectangle goes within a larger one, along one axis.

Used by RectangleOps.

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ShapeLocator group

Refers to Point, Rectangle (page 1).

Color color.ts

An sRGB color with straight (not premultiplied) alpha.

Used by InkStroke, Circle.

```
readonly r: number
readonly g: number
readonly b: number
readonly a: number
```

Shape shape.ts

A region of the plane, such as an ink stroke.

Used by InkStroke, Circle, ShapeLocator; AddShape (page 3); ShapeCollection, ShapeCollectionWriter, ShapeMessageProcessor (page 6).

```
readonly kind: string
```
Identifies the shape's type.
```
boundingBox(): Rectangle
```
The exact extents of the shape: x and y are its minimum x and y, and x + width and y + height are its maximum x and y.
```
intersects(r: Rectangle): boolean
```
Whether at least one point lies both in this shape and in r.
```
contains(p: Point): boolean
```
Whether p lies in the shape.

InkStroke extends Shape shape.ts

A stroke of digital ink: a polyline drawn with a round pen.

Extends Shape.

```
readonly kind: "ink"
readonly points: readonly Point[]
```
The polyline's vertices, in drawing order.
```
readonly thickness: number
```
Pen diameter, in canonical units.
```
readonly color: Color
```

Circle extends Shape shape.ts

A filled circle: the points within radius of center, with the half-open rule at its maximum x and y.

Extends Shape.

```
readonly kind: "circle"
readonly center: Point
readonly radius: number
```
In canonical units.
```
readonly color: Color
```

ShapeLocatorQueries shape.ts

Finds shapes by location.

Used by ShapeLocator; ShapeCollection (page 6).

```
inRect(r: Rectangle, out: ShapeId[]): number
```
Finds the shapes that intersect r.
```
atPoint(p: Point, out: ShapeId[]): number
```
Finds the shapes that contain p.

ShapeLocator extends ShapeLocatorQueries shape.ts

A spatial index that finds shapes by location, such as a quadtree.

Extends ShapeLocatorQueries.

```
add(id: ShapeId, shape: Shape): void
```
Indexes shape under id.
```
remove(id: ShapeId): boolean
```
Removes id from the index.
```
removeMultiple(ids: readonly ShapeId[]): number
```
Removes every listed ID that is present. ids may contain duplicates and IDs that aren't present; both are ignored.

Types

```
type ShapeId = string & { readonly __brand: "ShapeId" }
```
Identifies a shape on every client.

Used by ShapeLocatorQueries, ShapeLocator, ShapeId; AddShape, RemoveShapes (page 3); ShapeCollection, ShapeCollectionWriter, ShapeMessageProcessor (page 6).

Interface summary

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Operation group

Refers to ShapeId, Shape (page 2).

Operation dataModel.ts

One change to a data model.

Used by AddShape, RemoveShapes, ClearShapes; LocalQueue, MessageProcessor (page 5).

```
readonly type: string
readonly clientId: string
readonly clientSeq: number
readonly refSeq: number
seq: number
```

RemoveShapes extends `Operation` shapeModel.ts

Removes the listed shapes that are present.

Extends Operation.

```
readonly type: "remove"
readonly ids: readonly ShapeId[]
```

ClearShapes extends `Operation` shapeModel.ts

Removes the shapes its client knew about (refSeq in dataModel.ts).

Extends Operation.

```
readonly type: "clear"
```

AddShape extends `Operation` shapeModel.ts

Adds shape under id.

Extends Operation.

```
readonly type: "add"
readonly id: ShapeId
readonly shape: Shape
```

LayoutObject group

Refers to Rectangle, Point (page 1).

LayoutObject layout.ts

Something placed in a region of a window.

Used by EventDistributor.

```
region: Rectangle
```

Where it is, in the window's coordinates.

EventDistributor<M> extends `LayoutObject` layout.ts

Passes a platform's positional events to listeners by where they fall.

Extends LayoutObject.

```
locate(event: M[keyof M]): Point
```

Where event happened, in the window's coordinates.

```
on<E extends keyof M & string>(eventName: E, listener:
   Listener<M, E>, target: LayoutObject): void
```

Registers listener for eventName on target.

```
off<E extends keyof M & string>(eventName: E, listener:
   Listener<M, E>, target: LayoutObject): void
```

Removes the most recent registration of listener for eventName on target; does nothing if there is none.

```
distribute<E extends keyof M & string>(eventName: E,
   event: M[E], except?: LayoutObject): boolean
```

Passes event to the innermost target, among those registered for eventName other than except, that contains it.

Types

```
type Listener<M, E extends keyof M> = (eventName: E,
   event: M[E]) => void
```

Gets one of a platform's events: its name, and the platform's event object.

Used by EventDistributor.

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View group

Refers to Viewport (page 1); ShapeCollection (page 6).

EventEmitter<Events extends EventMap> events.ts

Something that emits events.

Used by View; DataModelReader (page 5).

```
on<E extends keyof Events & string>(event: E, listener:
    (...args: Events[E]) => void): this
```

Registers listener for event.

```
off<E extends keyof Events & string>(event: E, listener:
    (...args: Events[E]) => void): this
```

Removes the most recent registration of listener for event; does nothing if it has none.

FrameHandler frames.ts

Something that does its work during a display frame.

Used by FrameScheduler, View.

```
frame(timeMs: number): void
```

timeMs is the frame's timestamp in milliseconds, on the window's monotonic clock.

View<M extends ViewModel> extends FrameHandler, EventEmitter<{ viewport: [Viewport] }> view.ts

Shows one model through a viewport, and stays current.

Extends FrameHandler, EventEmitter. **Used by** ShapeView.

```
readonly model: M
viewport: Viewport
```

Setting this redraws the view at the next frame and emits viewport with the new value.

```
dispose(): void
```

Ends the view: removes its listener from the model, and requests, draws and emits nothing more.

ShapeView extends View<ShapeCollection> shapeView.ts

A view of shapes.

Extends View.

```
frame(timeMs: number): void
```

Draws the model as it is now.

FrameScheduler frames.ts

Calls handlers before the window's next display refresh.

```
requestFrame(handler: FrameHandler): void
```

Asks for one call to handler.frame() before the next display refresh.

Types

```
type EventMap = Record<string, unknown[]>
```

Each event's name, mapped to the arguments its listeners get.

Used by EventEmitter; ModelChangeEvents (page 5).

```
type ViewModel = DataModelReader<{ change: [] }>
```

A model a view can show: its read side, which emits change after each change.

Used by View.

Interface summary

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OperationSink group

Refers to Operation (page 3); EventMap, EventEmitter (page 4).

ModelChangeEvents extends `EventMap` dataModel.ts

A model's change events.

Used by DataModelReader.

DataModelReader`<E extends ModelChangeEvents>` extends `EventEmitter<E>` dataModel.ts

The read side of a data model.

Extends EventEmitter (page 4). **Used by** MessageProcessor; ShapeCollection (page 6).

OperationSink`<T>` dataModel.ts

Where producers put operations.

Used by OperationQueue, MessageProcessor.

`enqueue(item: T): void`

OperationQueue`<T>` extends `OperationSink<T>` dataModel.ts

A queue of operations, read by the processor.

Extends OperationSink.

`dequeue(): T | undefined`

The oldest item, removed; undefined when the queue is empty.

LocalQueue`<TOp extends Operation>` dataModel.ts

A processor's local queue. enqueue stamps the operation and returns it, so that the infrastructure can send it to collaborators; that is the only way a local operation leaves the processor.

Used by MessageProcessor.

`enqueue(content: OperationContent<TOp>): TOp`

Stamps content (MessageProcessor.stamps), queues it, and returns the stamped operation.

MessageProcessor`<TR extends DataModelReader, TOp extends Operation>` dataModel.ts

Applies operations to a data model: the only thing that changes it.

Used by ShapeMessageProcessor (page 6).

`readonly clientId: string`

The client whose local operations this processor applies.

`readonly model: TR`

The model's read side.

MessageProcessor, continued

`readonly local: LocalQueue<TOp>`

Where the infrastructure puts the operations controllers send.

`readonly remote: OperationSink<TOp>`

Where the collaboration protocol puts sequenced operations.

`process(): Promise<void>`

Applies the queued operations, interleaving local and remote, and emits the model's change events.

Types

```
type OperationContent<T extends Operation> = T extends
   Operation ? Omit<T, Exclude<keyof Operation, "type">>
   : never
```

What a controller forms: an operation without the fields the processor stamps on it.

Used by LocalQueue, LocalSend.

```
type LocalSend<TOp extends Operation> = (content:
   OperationContent<TOp>) => void
```

How a controller changes a model: it sends each local operation it forms to the message infrastructure.

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ShapeCollectionWriter group

Refers to ShapeLocatorQueries, Shape, ShapeId (page 2); DataModelReader, MessageProcessor (page 5).

DataModel dataModel.ts

The write side of a data model: modification methods in derived interfaces.

Used by ShapeCollectionWriter.

ShapeCollection<S extends Shape> extends DataModelReader<ShapeCollectionEvents>, ShapeLocatorQueries shapeModel.ts

Shapes in z-order, found by location.

Extends ShapeLocatorQueries (page 2); DataModelReader (page 5). **Used by** ShapeCollectionWriter, ShapeMessageProcessor; ShapeView (page 4).

`get(id: ShapeId): S | undefined`
The shape with this ID, or undefined if no such shape exists.

`readonly size: number`
The number of shapes.

`forEach(visit: (id: ShapeId, shape: S) => void): void`
Calls visit for each shape, bottom to top. visit must not change the collection.

`compareZ(a: ShapeId, b: ShapeId): number`
Negative if a is below b, positive if above.

ShapeCollectionWriter<S extends Shape> extends ShapeCollection<S>, DataModel shapeModel.ts

The write side.

Extends ShapeCollection, DataModel.

`add(id: ShapeId, shape: S, below?: ShapeId): void`
Adds shape under id: directly below the shape below, or on top when below is undefined.

`remove(ids: readonly ShapeId[]): number`
Removes every listed shape that is present; repeats and absent IDs are ignored.

ShapeMessageProcessor<S extends Shape> extends MessageProcessor<ShapeCollection<S>, ShapeOp> shapeModel.ts

Applies ShapeOps to a ShapeCollection through its writer.

Extends MessageProcessor (page 5).

`newShapeId(): ShapeId`
An ID for a new shape, for an AddShape this client makes.

Types

`type ShapeCollectionEvents = { change: [] }`
Used by ShapeCollection.

`type ShapeOp = AddShape | RemoveShapes | ClearShapes`
Used by ShapeMessageProcessor.